

Jupyter Notebook Installation Guide

For students and AI agents installing Jupyter on Windows, macOS, Linux, and VS Code
Prepared for the Data Analytics Project - checked against official sources on 12 June 2026

Purpose

This PDF gives exact, no-assumption installation instructions for Jupyter Notebook. It is written so a student can follow it manually or paste the included prompts into an AI agent and ask the agent to guide the setup step by step. It also explains how to open and run Analysis.ipynb inside Visual Studio Code.

Important: Jupyter Notebook needs Python. For beginners, Anaconda is the simplest path because it includes Python, Jupyter Notebook, and common data science packages. For users who already have Python, the official Jupyter pip path is also included.

1. Choose the Correct Installation Path

Situation	Use This Path	Why
Beginner, course project, no Python setup yet	Path A: Anaconda Navigator	Simplest. Installs Python, Jupyter Notebook, and many data science packages together.
Already comfortable with terminal and Python	Path B: Python + pip	Lightweight. Uses official Python and pip commands.
Linux user with system package manager or existing conda	Path B or Path C	Use Python + pip, or conda if already installed.
Wants to work inside VS Code	Install Jupyter first, then Path D: VS Code	VS Code opens ipynb files, but it still needs a Python environment with Jupyter installed.
Unsure	Path A, then Path D if using VS Code	Least assumptions for classroom data analytics work.

If an AI agent is helping, it must first identify the operating system, whether Python or Anaconda is already installed, whether VS Code is already installed, and whether the student is allowed to install software. It should not guess these details.

2. Pre-Install Checklist

Before installing anything, complete these checks.

- Confirm operating system: Windows, macOS, or Linux.
- Confirm processor type when needed: Intel/AMD 64-bit or Apple Silicon/ARM.
- Confirm internet access.
- Confirm whether the device is personal, university-managed, or company-managed.
- Confirm whether admin permission is available. If not, use user-level installation options.
- Confirm whether the student wants to use browser Jupyter Notebook, VS Code, or both.
- Close old Jupyter, Python, Anaconda, VS Code, or terminal windows before testing new commands.

3. Prompt to Give an AI Agent

Copy this prompt into an AI agent if you want it to guide the installation.

Help me install Jupyter Notebook for a data analytics class and help me run Analysis.ipynb. Do not assume my operating system or setup. First ask me what operating system I use, whether I already have Python or Anaconda installed, whether I already have VS Code installed, and whether I have admin permission. Then guide me using official Jupyter, Python, Anaconda, conda, and VS Code instructions. After installation, verify that Jupyter launches, then show me how to open the project folder and run Analysis.ipynb in both browser Jupyter and VS Code if I want VS Code. Do not skip verification steps.

4. Path A - Install with Anaconda Navigator

Recommended for beginners and most student teams. Project Jupyter documentation recommends Anaconda for new users because it installs Python, Jupyter Notebook, and common scientific packages together.

Windows

- Go to <https://www.anaconda.com/download> or the Anaconda installation documentation.
- Download the Windows graphical installer for your device.
- Run the installer. If unsure, accept default options.
- Open Anaconda Navigator from the Start menu.
- Launch Jupyter Notebook from Anaconda Navigator.
- In Jupyter, browse to the project folder and open Analysis.ipynb.

macOS

- Go to <https://www.anaconda.com/download> or the Anaconda installation documentation.
- Download the macOS graphical installer. Choose Apple Silicon if using M1/M2/M3/M4, or Intel if using an Intel Mac.
- Open the .pkg installer and follow the prompts.
- Open Anaconda Navigator from Applications or Launchpad.
- Launch Jupyter Notebook and open Analysis.ipynb from the project folder.

Linux

- Use the Anaconda Linux terminal installer from the official Anaconda installation documentation.
- Choose the installer matching your architecture, such as x86_64 or ARM64.
- Run the installer from Terminal and follow the prompts.
- Open a new Terminal after installation.
- Run `jupyter notebook` from the project folder.

```
# Linux example after Anaconda is installed
cd /path/to/Data_Analytics_Project
jupyter notebook
```

5. Path B - Install with Python and pip

Use this path if Python is already installed or if the student prefers a lighter setup. Official Jupyter installation instructions list `pip install notebook` for classic Jupyter Notebook and `jupyter notebook` to run it.

Step B1 - Check Python

```
# Windows
py --version
py -m pip --version

# macOS or Linux
python3 --version
python3 -m pip --version
```

If Python is missing, install it from <https://www.python.org/downloads/>. As of 12 June 2026, Python.org lists Python 3.14.6 as the latest source release. If a package compatibility issue appears, use Anaconda or a supported earlier Python 3 release accepted by the course.

Step B2 - Upgrade pip

```
# Windows
py -m pip install --upgrade pip

# macOS or Linux
python3 -m pip install --upgrade pip
```

Step B3 - Install Jupyter Notebook

```
# Windows
```

```
py -m pip install notebook

# macOS or Linux
python3 -m pip install notebook
```

Step B4 - Install common data analytics packages

For this project, the notebook will likely need pandas, NumPy, Matplotlib, Seaborn, and scikit-learn. Install them after Jupyter.

```
# Windows
py -m pip install pandas numpy matplotlib seaborn scikit-learn

# macOS or Linux
python3 -m pip install pandas numpy matplotlib seaborn scikit-learn
```

Step B5 - Launch Jupyter Notebook from the project folder

```
# Windows PowerShell example
cd "C:\Degree\A252\Data Analytics\Data_Analytics_Project"
py -m notebook

# macOS or Linux example
cd "/path/to/Data_Analytics_Project"
python3 -m notebook
```

A browser tab should open. If not, copy the local URL printed in the terminal, usually beginning with `http://localhost:8888/`, and paste it into a browser.

6. Path C - Install with conda

Use this path if conda is already installed through Anaconda, Miniconda, or Miniforge. The conda documentation supports Windows, macOS, and Linux, and states that admin permission is not required if installing into a user-writable location.

```
# Create a clean environment for the class project
conda create -n data-analytics python=3.12 -y
conda activate data-analytics

# Install notebook and common data analytics packages
conda install -c conda-forge notebook pandas numpy matplotlib seaborn scikit-learn -y

# Launch from the project folder
cd "path/to/Data_Analytics_Project"
jupyter notebook
```

The python=3.12 version is used here as a conservative classroom environment choice. If the instructor specifies another Python version, use that version instead.

7. Verify the Basic Jupyter Installation

Do not consider installation complete until all checks pass.

Check	Command or Action	Expected Result
Python works	<code>py --version</code> or <code>python3 --version</code>	Shows Python 3.x, not Python 2.
pip works	<code>py -m pip --version</code> or <code>python3 -m pip --version</code>	Shows pip version and Python location.
Jupyter command works	<code>jupyter notebook --version</code>	Shows a notebook version.
Notebook opens	<code>jupyter notebook</code> or <code>py -m notebook</code>	Browser opens Jupyter file browser.
Project opens	Open <code>Analysis.ipynb</code>	Notebook opens without file path errors.
Core packages import	Run import test cell	No import errors.

```
# Test cell inside Analysis.ipynb
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn

print("Setup OK")
```

8. Path D - Open and Run Analysis.ipynb in VS Code

Use this section if the student wants to work inside Visual Studio Code instead of the browser-based Jupyter interface. VS Code can open .ipynb files directly, but it still needs a Python or Anaconda environment where the Jupyter package is installed.

Step D1 - Install VS Code

- Go to <https://code.visualstudio.com/download>.
- Download the installer for Windows, macOS, or Linux.
- Install VS Code using the default options unless the device has university restrictions.
- Open VS Code after installation.

Step D2 - Install Required VS Code Extensions

- Open the Extensions panel in VS Code.
- Search for Python by Microsoft and install it.
- Search for Jupyter by Microsoft and install it if it is not already installed automatically.
- Reload VS Code if it asks you to reload.

Step D3 - Open the Project Folder

- In VS Code, choose File > Open Folder.
- Select the Data_Analytics_Project folder, not only the notebook file.
- If VS Code asks whether you trust the folder, choose Trust only if this is your project folder from the class repository.
- Confirm that Analysis.ipynb and the CSV file are visible in the Explorer panel.

Step D4 - Select the Python Interpreter

- Open the Command Palette: Ctrl+Shift+P on Windows/Linux or Cmd+Shift+P on macOS.
- Run Python: Select Interpreter.
- Choose the Python or conda environment where notebook, pandas, numpy, matplotlib, seaborn, and scikit-learn were installed.
- If using conda, choose the data-analytics environment if that was created earlier.

Step D5 - Open the Notebook and Select a Kernel

- Click Analysis.ipynb in the VS Code Explorer.
- If the notebook opens as plain text, confirm the file extension is .ipynb and that the Jupyter extension is installed and enabled.
- Click Select Kernel in the top-right corner of the notebook.
- Choose the same Python or conda environment selected in Step D4.
- If VS Code asks to install ipykernel or Jupyter dependencies in the selected environment, allow it only for that selected project environment.

Step D6 - Run the Notebook

- Run the first setup/import cell.
- If the first cell succeeds, use Run All only when you are ready to run the whole notebook.
- For team work, each person should run cells from the top through their assigned section.
- Save the notebook after adding your section and handoff note.
- Do not run destructive commands or delete another member's section.

Step D7 - VS Code Verification Cell

Run this in Analysis.ipynb inside VS Code.

```
import sys
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn

print("VS Code notebook setup OK")
print(sys.executable)
```

Prompt for an AI Agent to Help with VS Code

Help me open and run Analysis.ipynb in VS Code. Do not assume my setup. First check whether VS Code is installed, whether the Python and Jupyter extensions are installed, and which Python or conda environment has Jupyter and the data analytics packages. Then guide me to open the Data_Analytics_Project folder, trust the workspace only if it is the correct class folder, select the Python interpreter, select the notebook kernel, run the import test cell, and then run Analysis.ipynb. If anything fails, explain the exact error and fix only that issue.

9. Troubleshooting

Problem	Likely Cause	What to Do
jupyter is not recognized	Jupyter installed but not on PATH, or not installed in active environment.	Try <code>py -m notebook</code> on Windows or <code>python3 -m notebook</code> on macOS/Linux. If using conda, run <code>conda activate data-analytics</code> first.
pip is not recognized	pip command not on PATH.	Use <code>py -m pip</code> on Windows or <code>python3 -m pip</code> on macOS/Linux.
Browser does not open	Jupyter server started but browser launch failed.	Copy the local URL from terminal and paste it into a browser.
Port 8888 is busy	Another Jupyter server is running.	Use the alternate URL Jupyter prints, or run <code>jupyter notebook --port 8890</code> .
Package import fails	Package missing from environment.	Install the missing package in the same environment used to launch Jupyter.
Permission error during install	No admin rights or protected install location.	Use Anaconda user install, conda user-writable location, or ask IT/admin for help.
Notebook opens in wrong folder	Jupyter launched from another directory.	Close Jupyter, <code>cd</code> into the project folder, then launch again.
VS Code cannot run cells	No kernel selected, untrusted workspace, or missing Jupyter extension.	Trust the correct project folder, install the Jupyter extension, and select a kernel from the notebook top-right picker.
VS Code imports fail but terminal works	VS Code is using a different Python interpreter or kernel.	Use Python: Select Interpreter and Select Kernel to choose the environment where packages were installed.
Analysis.ipynb opens as text	Jupyter extension missing or disabled, or file extension is wrong.	Install/enable the Jupyter extension and confirm the file name ends with <code>.ipynb</code> .

10. AI Agent Safety Checklist

- The agent must ask which operating system is being used before giving commands.
- The agent must ask whether the student wants browser Jupyter, VS Code, or both.
- The agent must not mix Anaconda, conda, and pip commands unless it explains which environment is active.
- The agent must verify installation with version commands and an import test.
- The agent must launch Jupyter or VS Code from the project folder that contains Analysis.ipynb.
- The agent must not delete existing project files.
- The agent must stop and ask for help if admin permission, antivirus, or university device policy blocks installation.

11. Quick Commands Summary

Windows with Python + pip

```

py --version
py -m pip install --upgrade pip
py -m pip install notebook pandas numpy matplotlib seaborn scikit-learn
cd "C:\Degree\A252\Data Analytics\Data_Analytics_Project"
py -m notebook

```

macOS or Linux with Python + pip

```

python3 --version
python3 -m pip install --upgrade pip
python3 -m pip install notebook pandas numpy matplotlib seaborn scikit-learn
cd "/path/to/Data_Analytics_Project"
python3 -m notebook

```

Any OS with conda

```

conda create -n data-analytics python=3.12 -y
conda activate data-analytics
conda install -c conda-forge notebook pandas numpy matplotlib seaborn scikit-learn -y
cd "path/to/Data_Analytics_Project"
jupyter notebook

```

VS Code quick path

- Install VS Code.
- Install Microsoft Python and Jupyter extensions.
- Open the Data_Analytics_Project folder.
- Run Python: Select Interpreter.
- Open Analysis.ipynb.
- Use Select Kernel in the top-right of the notebook.
- Run the import test cell, then run the notebook.

12. Official Sources Checked

These sources were checked on 12 June 2026. If instructions change later, prefer the official pages below.

Source	URL
Project Jupyter install page	https://jupyter.org/install
Jupyter documentation - classic notebook install	https://docs.jupyter.org/en/latest/install/notebook-classic.html
Python downloads	https://www.python.org/downloads/
conda installation documentation	https://docs.conda.io/projects/conda/en/latest/user-guide/install/index.html
Anaconda installation documentation	https://www.anaconda.com/docs/getting-started/anaconda/install/overview
JupyterLab installation documentation	https://jupyterlab.readthedocs.io/en/stable/getting_started/installation.html
VS Code Jupyter notebooks documentation	https://code.visualstudio.com/docs/datascience/jupyter-notebooks
VS Code Jupyter kernel management documentation	https://code.visualstudio.com/docs/datascience/jupyter-kernel-management

Final Instruction for Students

After setup, open Analysis.ipynb either in browser Jupyter or in VS Code. Run the notebook from the top. Each team member should complete only their assigned section and leave a short Markdown handoff note for the next member.